



Sanjivani Rural Education Society's
Sanjivani College of Engineering, Kopargaon-423 603
(An Autonomous Institute, Affiliated to Savitribai Phule Pune University, Pune)
NAAC 'A' Grade Accredited, ISO 9001:2015 Certified

Department of Computer Engineering

(NBA Accredited)

Subject- Object Oriented Programming (CO212)
Unit 1 – Fundamentals of OOP
Topic –1.10 Functions in C++

Prof. V.N.Nirgude
Assistant Professor
E-mail : nirgudevikascomp
@sanjivani.org.in
Contact No: 9975240215



C++ Functions

- C++ allows the use of both internal (user-defined) and external (built-in) functions.
- External functions are usually grouped into specialized libraries (e.g., iostream, stdlib, math, etc.)



Function Definition

A set of statements that explains what a function does is called Function Definition

Return-type Function-name (parameters)

Function header

```
{  
    statement 1;  
    statement 2;  
    :  
    :  
    :  
    statement N;  
}
```

Function body



Function Declaration

It is also known as FUNCTION PROTOTYPE.

It provides information to compiler about the structure of function to be used in program.

It ends with semicolon (;).

It consists of:

- FUNCTION NAME
- FUNCTION RETURN TYPE
- NUMBERS & TYPES OF PARAMETERS



Syntax of Function Declaration

Return-type Function-name (parameters);

**Indicates the type
of value that will
be returned by
function**

**Indicates the
name of
function**

**Parameters are
the values that
are provided to a
function when the
function is called.**

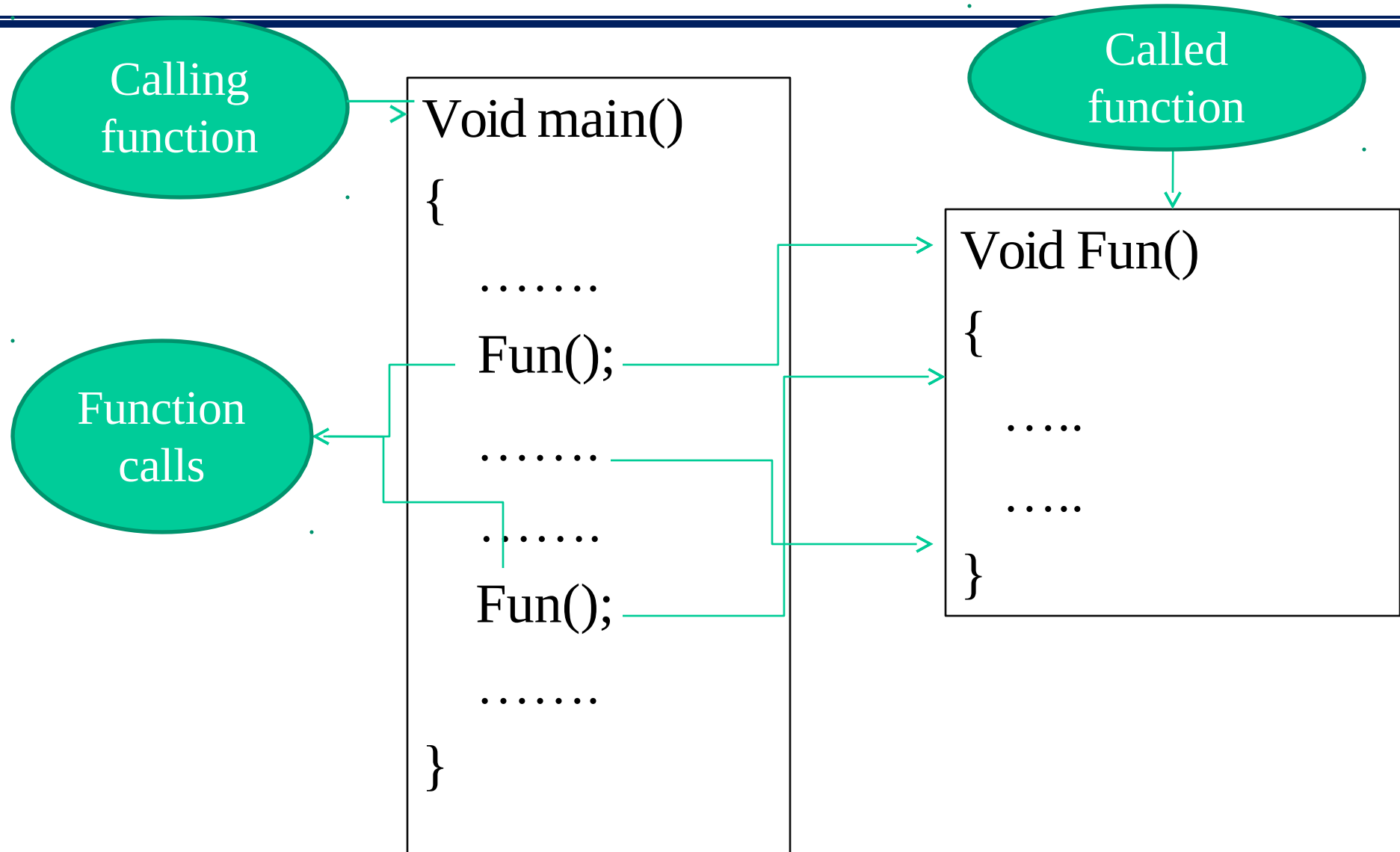


Function Call

- The statement that activates a function is known as FUNCTION CALL.
- The following steps take place when a function is called:
 1. The control moves to the function that is called.
 2. All statements in function body are executed.
 3. Control returns back to calling function.



Function Call Mechanism





Example using Function

```
1  #include<iostream>
2  using namespace std;
3
4  int sum (int x , int y);           // function declaration
5  int sum(int x , int y)           // funciotn definition
6  {
7      int result;
8      result = x + y;
9      return (result);
10 }
11 int main()
12 {
13     int x , y , output ;
14     x = 20;
15     y = 100;
16     output = sum(x,y);             /* calling a function and storing the
17                                     value from funciton to variable output*/
18     cout<<output;
19
20     return 0;
21 }
```




Utility Functions

- C++ has several utility functions for number of conversions, argument processing, sorting and searching and random number generation. To use these function the header `<cstdlib>` is required.
- The following table lists the important utility functions of C++ Standard Function Library.

Functions	Description
<u>abort</u>	Terminates the program abnormally.
<u>abs</u>	Returns the absolute value of the number.
<u>bsearch</u>	Performs a binary search on an array.
<u>exit</u>	Causes normal and immediate termination of a program.
<u>qsort</u>	Function is used to quick sort an array.
<u>rand</u>	Generates a random number.

Thank you..